

IndigoSCADA: A Small Footprint SCADA System

ANKITA K.S.

Processes like controlling, supervising, collecting and analysing data and generating reports are crucial for any industry. It is always wise to automate these processes. For the same, there are a number of such control system software available, but supervisory control and data acquisition (SCADA) tools have always been reliable. One such SCADA system is IndigoSCADA.

Noteworthy traits of IndigoSCADA

IndigoSCADA is a powerful SCADA tool used in small custom projects where MODBUS, DNP3, OPC DA 2.05, OPC A&E 1.1, IEC 60870-5-101, IEC 60870-5-103 and IEC 60870-5-104 are used. There are a lot of special features that make IndigoSCADA useful when compared to other SCADA software.

The deal with data. IndigoSCADA uses data to generate daily, weekly or monthly management of reports and graphically presents both historical and real-time data. It consists of an inbuilt structured query language (SQL) editor that allows for online maintenance of real-time and historical databases. In this system, resources such as people and plant assets are optimised, thus, promising high-level control over the plant environment.

Amenities by general-purpose licences. A key feature of IndigoSCADA is that it is an open source software, distributed under GNU lesser general-public licence, and some of the drivers under non-FLOSS licence. However, being open source, the whole software is well grounded because of the number of iterations and revisions

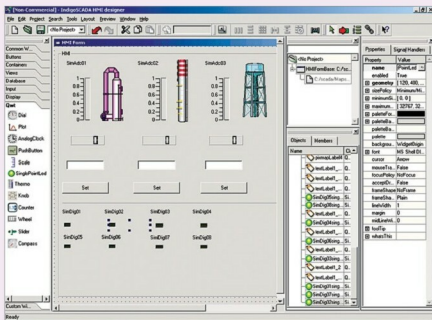


Fig. 1: IndigoSCADA HMI designer

Overview of SCADA

SCADA is an industrial control system where a computer system monitors and controls a process. A SCADA system coordinates but does not control the process in real time. To perform real-time operations, it often includes distributed control system components.

A typical SCADA system consists of five main components. These are HMI, supervisory system, remote terminal unit, programmable logic controller and communication infrastructure.

Functions of the components. HMI acts as an interface that presents processed data to a human operator. Whereas, a supervisory system is nothing but a computer that gathers data from the process and sends commands or control signals to the process. The main function of a remote terminal unit is to connect to sensors in the process, thereby converting sensor signals to digital data and sending digital data to the supervisory system.

There are special-purpose remote terminal units available too, but programmable logic controllers are preferable because these are more economical, versatile, flexible and configurable. When it comes to communication infrastructure, these provide connectivity between supervisory systems and remote terminal units.

it has gone through. Also, frequent developments in the code make constant updating of features possible.

The language boon. This

system is developed entirely in C and C++ with multiple operating system support and multiple front-end protocol drivers. This makes it